Linear regression: Abalone growth and pH example

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Table of contents

Set up

```
library(tidyverse) # general use
library(janitor) # cleaning data frames
library(here) # file/folder organization
library(readxl) # reading .xlsx files
library(ggeffects) # generating model predictions
library(flextable) # creating tables
```

```r
# abalone data from Hamilton et al. 2022
# here is a package/function that points computer to right folder in a directory
# read xlsx reads in excel files
abalone <- read_xlsx(here("data", "Abalone IMTA_growth and pH.xlsx"))
```

Abalone example

Data from Hamilton et al. 2022. "Integrated multi-trophic aquaculture mitigates the effects of ocean acidification: Seaweeds raise system pH and improve growth of juvenile abalone." https://doi.org/10.1016/j.aquaculture.2022.738571

a. Questions and hypotheses

Question: How does pH predict abalone growth (measured in change in shell surface area per day, mm^2 d^{-1})?

H_0: **There is no effect of pH on abalone growth.**

H_A: **There is an effect of pH on abalone growth.**

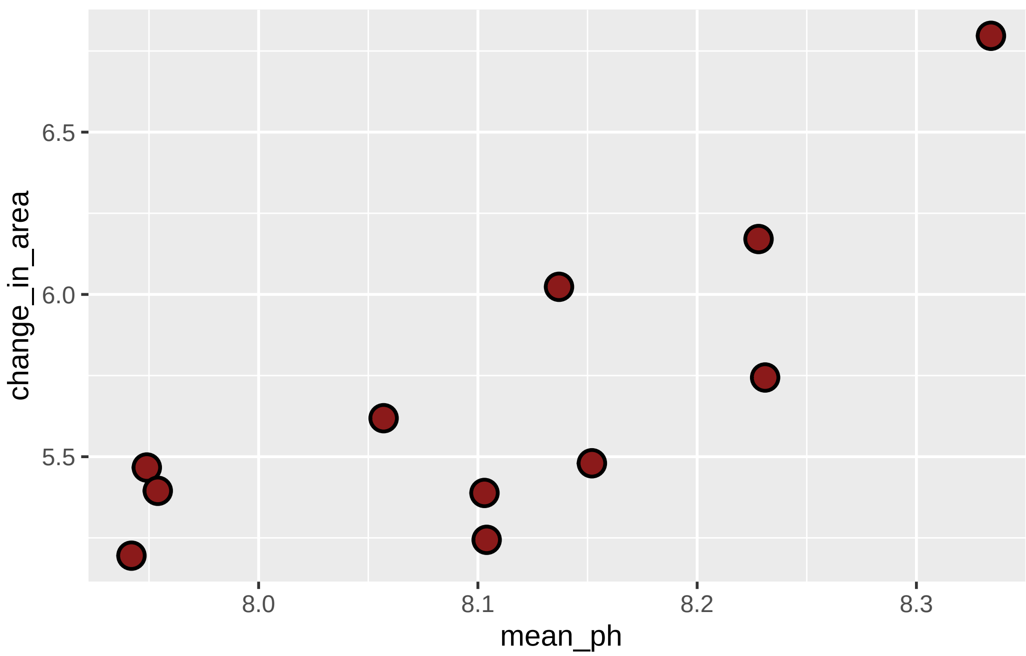
b. Cleaning

```r
# creating a new object from original data set
abalone_clean <- abalone  |>
  # cleaning column names
  clean_names() |>
  # selecting desired columns
  select(mean_p_h, change_in_area_mm2_d_1_25) |>
  # renaming columns for clarity
  rename(mean_ph = mean_p_h,
         change_in_area = change_in_area_mm2_d_1_25)
```

Don't forget to look at your data! Use `View(abalone_clean)` in the Console.

c. Exploratory data visualization

```
# base layer: ggplot
ggplot(data = abalone_clean, # read in abalone_clean data
       aes(x = mean_ph, # make x-axis mean_ph, y-axis change_in_area
           y = change_in_area)) +
  # first layer: geom_point to represent individual observations
  geom_point(size = 4,
             stroke = 1,
             fill = "firebrick4",
             shape = 21)
```



Where does the missing value come from?

The missing value comes froma. pair of tanks in which the pH sensor malfunctioned, so there is no pH reading for those tanks.

Don't forget to turn your warnings off either in the individual code chunk or in the YAML at the top of the file.

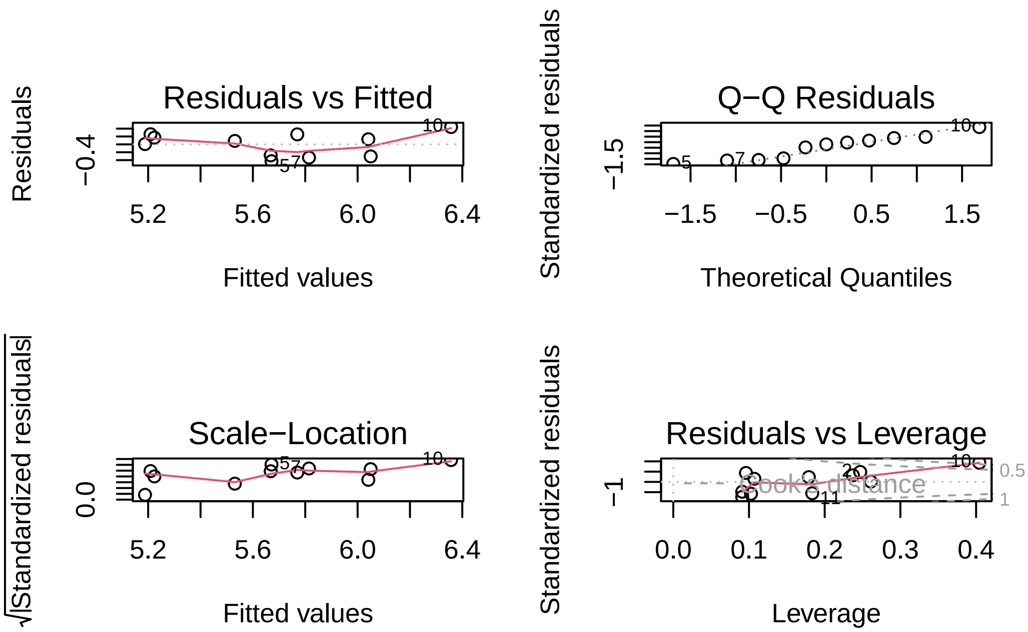
d. Abalone model

Model fitting

```
abalone_model <- lm(
  change_in_area ~ mean_ph, # formula: change in shell SA as a function of pH
  data = abalone_clean      # data frame: clean abalone data
)
```

Diagnostics

```
par(mfrow = c(2, 2))    # displays plots in a 2x2 grid
plot(abalone_model)     # display diagnostic plots
```



Model summary

```
# more information about the model
summary(abalone_model)
```

```
Call:
lm(formula = change_in_area ~ mean_ph, data = abalone_clean)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.42714 -0.29282  0.08769  0.21258  0.43965 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)   
(Intercept) -18.5010     6.1601  -3.003  0.01488 * 
mean_ph       2.9827     0.7596   3.926  0.00348 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3063 on 9 degrees of freedom
  (1 observation deleted due to missingness)
Multiple R-squared:  0.6314,	Adjusted R-squared:  0.5905 
F-statistic: 15.42 on 1 and 9 DF,  p-value: 0.003477
```

What is the slope estimate (with standard error?)

β_1 = 2.98 $\pm$ 0.76

What is the intercept estimate (with standard error?)

β_0 = -18.50 $\pm$ 6.16

Generating predictions

```r
# creating a new object called abalone_preds
abalone_preds <- ggpredict(
  abalone_model,      # model object
  terms = "mean_ph"   # predictor (in quotation marks)
)

# display the predictions
abalone_preds
```

```
# Predicted values of change_in_area

mean_ph | Predicted |     95% CI
```

```
--------------------------------
   7.94 |      5.19 | 4.83, 5.54
   7.95 |      5.21 | 4.86, 5.55
   8.06 |      5.53 | 5.30, 5.76
   8.10 |      5.67 | 5.46, 5.88
   8.10 |      5.67 | 5.46, 5.88
   8.14 |      5.77 | 5.55, 5.98
   8.15 |      5.81 | 5.59, 6.04
   8.33 |      6.36 | 5.92, 6.80
```

Look at the column names using `colnames(abalone_preds)` in the Console.

What are the column names?

`x`, `predicted`, `std.error`, `conf.low`, `group` (not going to worry about `group`)

Look at the actual object (by clicking on it in Environment).

Look at the class of the object using `class(abalone_preds)` in the Console.

What is this type of object?

This is a ggeffects object, but also a data frame.

View `abalone_preds` like a data frame.

Why generate model predictions?

This dataset does not include any actual observations of abalone growth at pH = 8.

However, the model can tell us a prediction for what abalone growth could be if pH = 8 based on the existing data.

Figure out what abalone growth the model would predict at pH = 8.

```
# finding the model prediction at a specific value
ggpredict(
  abalone_model,      # model object
  terms = "mean_ph[8]"   # predictor (in quotation marks) and predictor value in brackets
)
```

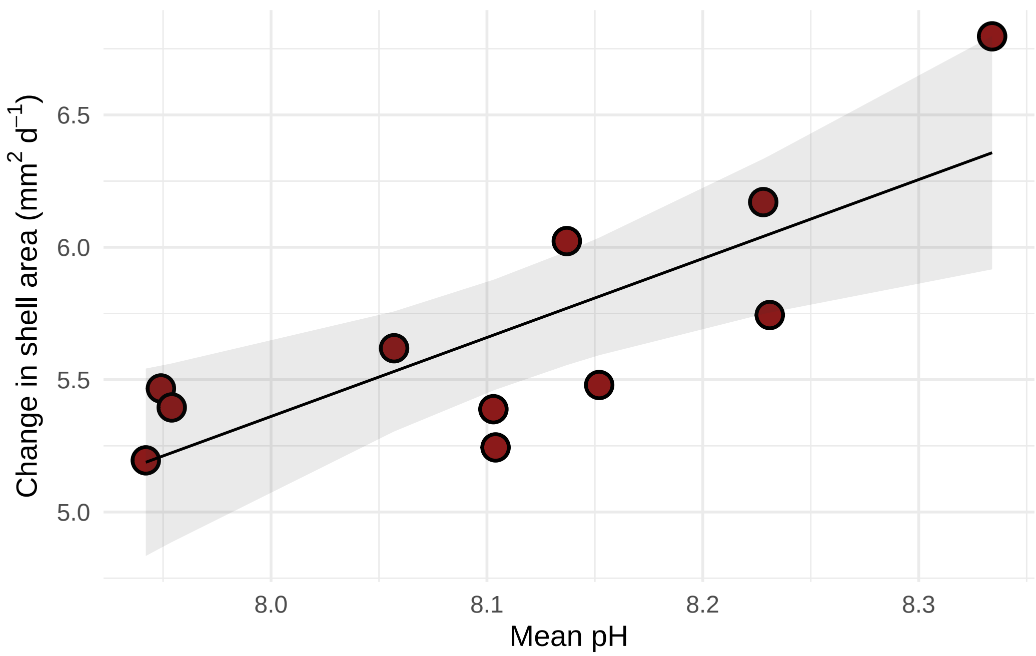
```
# Predicted values of change_in_area

mean_ph | Predicted |     95% CI
--------------------------------
      8 |      5.36 | 5.08, 5.64
```

How would you interpret this?

At pH = 8, the model predicts abalone shell growth of 5.36 [95% CI: 5.08, 5.64] mm^2 d^{-1}.

Visualizing model predictions and data

```
# base layer: ggplot
# using clean data frame
ggplot(data = abalone_clean,
       aes(x = mean_ph,
           y = change_in_area)) +
  # first layer: points representing abalones
  geom_point(size = 4,
             stroke = 1,
             fill = "firebrick4",
             shape = 21) +
  # second layer: ribbon representing confidence interval
  # using predictions data frame
  geom_ribbon(data = abalone_preds,
              aes(x = x,
                  y = predicted,
                  ymin = conf.low,
                  ymax = conf.high),
              alpha = 0.1) +
  # third layer: line representing model predictions
  # using predictions data frame
  geom_line(data = abalone_preds,
            aes(x = x,
                y = predicted)) +
  # axis labels
  labs(x = "Mean pH",
       y = expression("Change in shell area ("*mm^{2}~d^-1*")")) +
  # theme
  theme_minimal()
```



Creating a table with model coefficients, 95% confidence intervals, and more

Note: this function is from `flextable`.

```
# create table using as_flextable
as_flextable(abalone_model)
```

	Estimate	Standard Error	t value	Pr(>|t|)	
(Intercept)	-18.501	6.160	-3.003	0.0149	*
mean_ph	2.983	0.760	3.926	0.0035	**

*Signif. codes: 0 <= '***' < 0.001 < '**' < 0.01 < '*' < 0.05*

Residual standard error: 0.3063 on 9 degrees of freedom

Multiple R-squared: 0.6314, Adjusted R-squared: 0.5905

F-statistic: 15.42 on 9 and 1 DF, p-value: 0.0035

END OF ABALONE EXAMPLE